

Interview Questions — Database - No SQL- Mongodb

Total Questions: 45

1. What is NoSQL, and how is it different from traditional RDBMS databases?

Hint: Think about schema, scalability, and data structure flexibility.

2. Explain the structure of MongoDB. What are collections and documents?

Hint: Consider how data is organized and stored in MongoDB compared to tables and rows.

3. What is BSON in MongoDB, and why is it used instead of JSON?

Hint: Focus on the need for data types and efficiency.

4. How would you install MongoDB on your local machine?

Hint: Mention the general steps, not specific commands.

5. Describe the steps to create a new database and a collection in MongoDB.
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6. What is the difference between dropping a database and dropping a collection in MongoDB?

Hint: Think about the scope of data removal.

7. How do you create a new collection in MongoDB using the shell?
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8. Write the command to drop a collection named 'students' from the 'school' database in MongoDB.
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9. What are the main CRUD operations in MongoDB, and what does each one do?

Hint: List and briefly describe each operation.

10. How would you insert a single document and multiple documents into a collection using MongoDB shell commands?
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11. Write a MongoDB query to find all students in the 'students' collection who have scored more than 80 marks.

Hint: Use the appropriate query operator for comparison.

12. What is the difference between find() and findOne() in MongoDB?

13. How would you update the email address of a user with a specific user ID in MongoDB?

Hint: Mention the update operator you would use.

14. Write a MongoDB command to delete all documents from the 'orders' collection where the status is 'cancelled'.

15. Explain the difference between updateOne() and updateMany() in MongoDB.

16. What does the \$set operator do in MongoDB updates?

Hint: Think about modifying fields in a document.

17. How would you increment the 'quantity' field by 2 for a specific product in an inventory collection?

Hint: Which update operator helps with this?

18. Which operator would you use to remove a field from a document in MongoDB, and how does it work?

19. Write a query to add a new tag to a product's 'tags' array only if it does not already exist.

Hint: Consider the difference between \$push and \$addToSet.

20. What is the purpose of the \$match stage in the MongoDB aggregation framework?

Hint: Relate it to filtering data.

21. How would you use the aggregation pipeline to calculate the total sales amount from an 'orders' collection?

Hint: Think about \$group and sum.

22. Explain the difference between \$project and \$group in MongoDB aggregation.

23. Write an aggregation pipeline to get the average marks of students in each class from a 'students' collection.

24. What are indexes in MongoDB, and why are they important for performance?

Hint: Consider query speed and data retrieval.

25. Name two types of indexes available in MongoDB and explain when you would use each.

26. How would you create an index on the 'email' field of a 'users' collection?

Hint: Mention the shell command used.

27. What is the potential downside of having too many indexes on a MongoDB collection?

Hint: Think about write performance and storage.

28. How can you measure the impact of an index on query performance in MongoDB?

Hint: There's a command that shows query execution details.

29. How do you create a new user in MongoDB with read-only access to a specific database?

30. What is the purpose of roles in MongoDB's user management system?

31. Explain how authentication works in MongoDB and why it is important.

32. Describe a scenario where you would use an admin user versus a read-only user in MongoDB.

33. What is MongoDB Compass, and how does it help in database management?

Hint: Think about GUI vs CLI.

34. How would you deploy a MongoDB database on MongoDB Atlas?

35. Name one tool you can use to import data into MongoDB and explain how it works.

36. How can you export data from a MongoDB collection to a JSON or CSV file?

37. In an e-commerce project, how would you design the collections for products and orders in MongoDB?

Hint: Think about document structure and relationships.

38. Write a MongoDB aggregation query to generate a report of total sales per product category from an 'orders' collection.

39. How would you update the stock quantity for a product after an order is placed in your e-commerce database?

Hint: Consider an update operator and a practical workflow.

40. Describe how you would use MongoDB's aggregation framework to find the top 5 best-selling products.

41. How could you use an AI tool like ChatGPT to help you write or debug a MongoDB query, and what steps would you take to ensure the AI's suggestion is correct?

Hint: Think about verifying output and understanding the logic.

42. In what ways could AI-powered tools integrated into MongoDB Compass or Atlas assist you in managing your database, and where would you still need to rely on your own knowledge?

Hint: Consider suggestions, automation, and the importance of manual review.

43. Integrative: Explain how indexing can improve the performance of aggregation queries in MongoDB.

44. Integrative: In a student management system, how would you use both update operators and aggregation pipelines to maintain and report on student grades?

45. Integrative: Describe a workflow where you use MongoDB Atlas to deploy a database, set up user roles, and import initial data for an e-commerce mini project.
